# A Beginner's Guide to Open Source Security Tools

Open source security tools are as good as their commercial counterparts, if not better.


Here's a truth nobody told you: attackers don't care if you're a Fortune 500 company or a hobbyist with a VPS. Automated scans probe every IP address on the internet. Your server, your container, your laptop—they're all being checked for open doors right now.

Commercial security tools cost thousands. The open source alternatives cost nothing but your time. And they're good. Really good. Organisations with billion-dollar security budgets use the same tools you can download today.

Let's meet the tools…

## ClamAV: The antivirus that doesn't suck

This open source antivirus engine scans files for malware, viruses, and Trojans. So why is it different? Unlike bloated Windows antivirus suites that slow your machine to a crawl, ClamAV is lean. It runs on Linux, BSD, macOS, and yes, Windows. It works from the command line. It integrates with mail servers, file servers, and CI pipelines.

Use it for:

- Scanning file uploads on your web application
- Checking attachments on your mail server
- Verifying downloads before execution
- Adding malware detection to your CI/CD pipeline

This is how you can get started:

```
# Debian/Ubuntu
sudo apt install clamav clamav-daemon

# Update virus definitions
sudo freshclam

# Scan a directory
clamscan -r --infected /home/user/downloads

# Scan and remove threats (use carefully)
clamscan -r --remove /path/to/scan
```

*Pro tip:* Don't run real-time scanning on production servers unless you need it. It eats I/O. Use on-demand scanning for file uploads and scheduled scans for maintenance windows.

## Snort: The original intrusion detective

A network intrusion detection system (IDS), Snort watches your network traffic and raises alarms when it sees something suspicious.

Snort uses rules—thousands of them—to match patterns in network packets. Someone trying SQL injection? Snort sees the pattern. A port scan from a malicious IP? Snort logs it. Malware phoning home? Snort notices.

Use it for:

- Monitoring network traffic for attacks
- Detecting policy violations (P2P on corporate networks, etc)
- Forensic analysis after incidents
- Learning how attacks actually look on the wire

Here's a quick start:

```
# Ubuntu/Debian
sudo apt install snort

# Basic configuration
sudo snort -T -c /etc/snort/snort.conf

# Run in packet logging mode
sudo snort -dev -l /var/log/snort
```